

Application Research of Computer Artificial Intelligence Technology in Robot Multi-Function Sensor System

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Abstract—In this paper, a novel multichannel ranging method is proposed. The system consists of motion control, vision, ultrasound, laser, wireless communication and so on. It can be measured by different distance sensors according to different applications. The two types of distance information are synthetically processed by ultrasonic detection method. A combined periodic transmission method based on multiple ultrasonic sensors is proposed. It is proved by experiments that this new measuring method can achieve millimeter range in short distance measurement. The test system can solve the problem of collision avoidance and positioning of mobile manipulator in complex environment.

Keywords—Artificial intelligence, robot system, multifunctional sensor, ultrasonic sensor, group cyclic emission technology, data fusion

I. INTRODUCTION

The people pay more and more attention to its application in industrial production. It is an intelligent technology that integrates perception, decision making, behavior and interaction, and is an extension of human intelligence. This project integrates motion control, computer vision, path planning, wireless communication, information fusion and other disciplines, which can lay a foundation for its research and development and application [1]. Most of the control systems of many robots are

composed of special single-chip computers and special host boards. The system integrates the control algorithm and motor control into a microcontroller. The existing microcontroller processing and peripheral equipment resources are limited, it is difficult to expand the functional units of large information such as images, thus reducing the work efficiency of robots. This is not conducive to subsequent maintenance and function expansion. Open robots use a unified operating system and a standardized program development environment. Hardware connected with a standard bus architecture and sensors. Adopting the way of networking, the platform can easily realize the communication and information sharing with other systems. The architecture uses an efficient microcontroller, which can solve the complex computation problem well. A modular and hierarchical method is proposed to realize the overall performance of the system. Components can be flexibly matched for easy custom updates [2]. The mobile robot is placed in a 360-degree position to obtain more comprehensive environmental data. Multiple ultrasonic systems and a single ultrasonic probe will interfere with each other, resulting in multiple ultrasonic systems interfering with each other. The collected original data is preprocessed, and the evidence theory is used to achieve the accurate expression of various scene models and action selection.

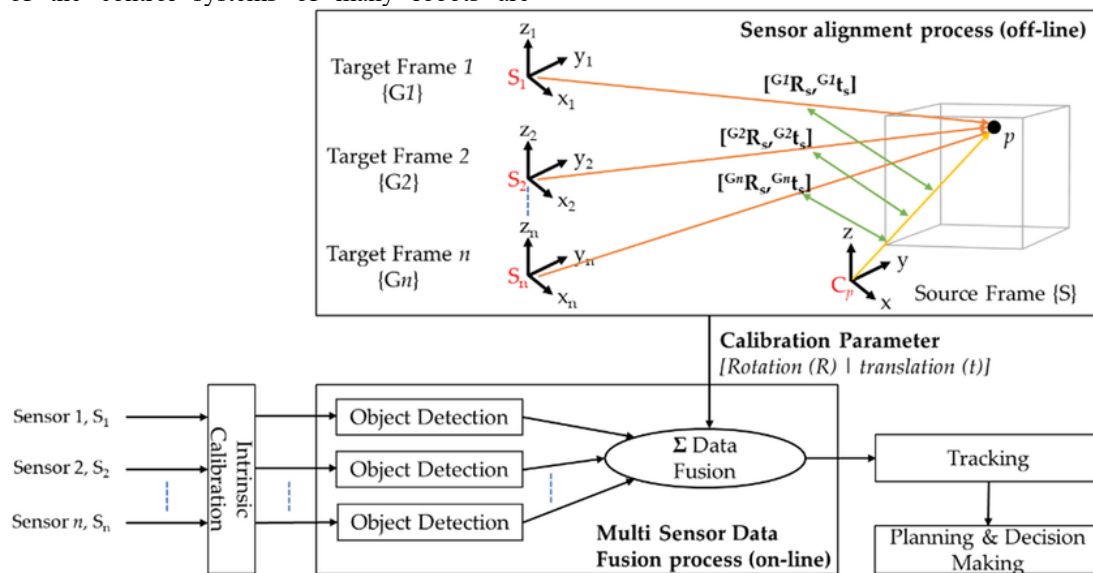


Fig. 1. MultiSensor data fusion grading system

II. MULTI-CHANNEL RADAR RANGE MEASUREMENT SYSTEM

A. Ranging system architecture

Figure 1 shows the multi-sensor information fusion hierarchy. At present, according to the level of information to be processed by fusion. The goal of the hierarchy is to enable different types of information to be treated differently during the Fusion process (Figure 1 cited in Sensor and Sensor Fusion Technology in Autonomous Vehicles: A Review).

"Power Storm" selects the 68HC11 series of microprocessors with the best performance and most complete configuration among 8-bit microprocessors. It has high stability and can be downloaded automatically. The composition of the control circuit is shown in Figure 2 (image cited in J. Mar. Sci. Eng. 2021, 9(9), 1020). Compared with previous industrial robots, this robot is smaller, its shape is similar to a space ship, its diameter is only 30 centimeters, and its mass is only 1 kilogram. This paper introduces a new intelligent system based on infrared

sensor, which uses three different sensors [3]. This sensor gives a "power storm" type sensor, which can sense the impact signal on the collision ring. Four (normally open) impact switches are installed on the left front, right front, left back and right back of the robot, and together with the impact ring, an impact sensor is formed. A receiving tube and a transmission tube are arranged on both sides of the robot's head. It sends and receives infrared rays of the occluded object to judge whether there are obstacles in front, front and right. The AS ultrasonic detector can emit ultrasonic waves, and then calculate the distance between targets by measuring the speed of the signal. Ultrasonic sensors can accurately detect targets ranging from 30 cm to 3 m. Using this method, a large area of the surrounding environment can be obtained. AS type infrared laser ranging system uses the phase difference between infrared radiation and reflected light to detect the target. AS type infrared ranging system is used to accurately detect obstacles in the range of 15-50 cm.

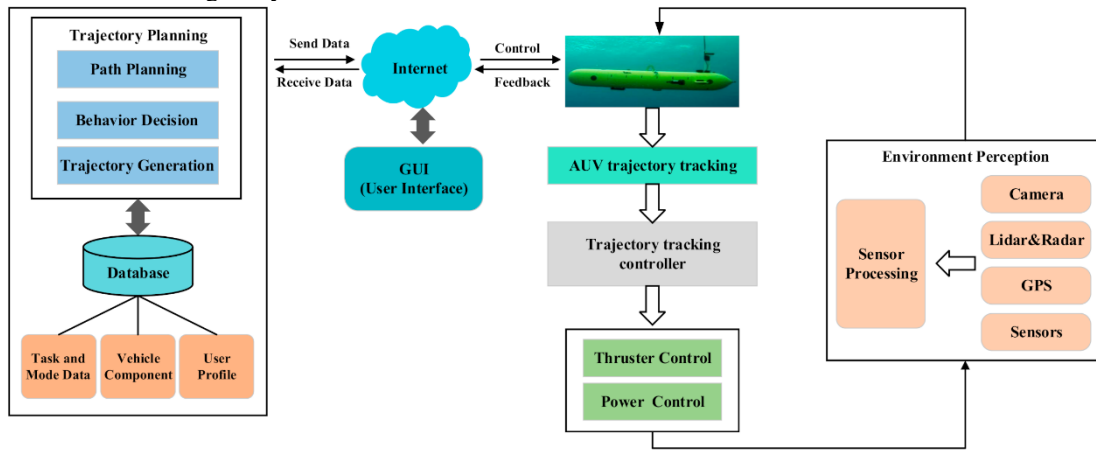


Fig. 2. Structure diagram of capability storm control circuit

B. Compound sensor

1) Ultrasonic distance measurement module

The main controller transmits the instructions to the MCU through serial communication, and the MCU activates each ultrasonic component according to the instructions to

realize the obstacle detection in all directions. The measured range data is fed back to the control center for judgment. The effective ultrasonic detection range is 0.5 m. Its structure is shown in Figure 3 (image cited in Sensors 2021, 21(6), 2241).

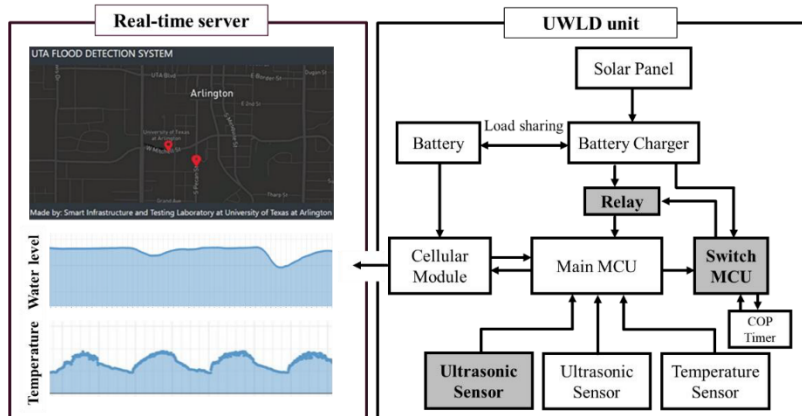


Fig. 3. Configuration of ultrasonic detection components

The input and output ports of the MCU emit a 9-channel square wave of 40 KHz. A frequency of 40 KHZ is

modulated with 74HC14M, and then the sensor is excited to generate an ultrasonic signal. The acquisition and analysis

of ultrasonic signal include: receiving transducer, amplifying, filtering, shaping, triggering output and so on. Because the energy of ultrasound decreases gradually with the increase of the propagation distance during the propagation process, it is necessary to carry out multi-stage signal amplification [4]. In the front end, NE5532 amplifier is used to amplify the received signal ten thousand times and filter the interference of other frequency bands. In the latter stage, the comparator of LM311 is connected to a voltage comparator in series. A useful signal is obtained by comparing the output results. The Circuit for Ultrasonic acceptance and processing is shown in Figure 4 (image cited in Best Ultrasonic Circuit Projects for Hobbyists and Engineers).

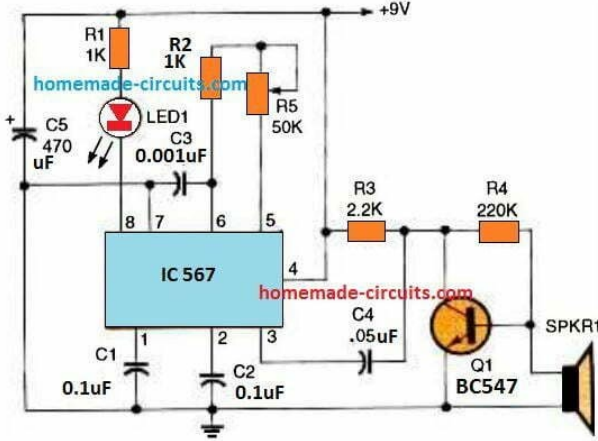


Fig. 4. Ultrasonic receiving circuit

2) Infrared distance measuring device

The infrared sensor uses the principle of triangulation to determine the distance. The light from the infrared transmitter hits the object and bounces back. The CCD detector detects and receives the reflected light. The distance D is then calculated based on the compensation value L , the emission Angle α , the center distance X , and the length f of the filter [5]. The system has the characteristics of fast response and low cost. At present, the main problems faced by infrared ranging technology are: the detection ability of absorbent material blocking or approaching black barrier is poor, and the infrared ranging module has the largest detection range.

3) Laser distance measuring device

LIDAR is a kind of laser radar that uses time-domain ranging method to achieve range. The exact distance is determined by the time difference between light and light. The system has the characteristics of high precision and wide application area. There are two ways to acquire time: one is based on pulse measurement and the other is based on phase measurement [6]. The pulse method is based on the diffusion characteristics of light, while the phase method is based on the phase change between the modulated light waves to infer the required time. Due to the role of meteorological factors and the "mirror-black hole" effect, if the detected target has good reflective performance, or its position is not consistent with the direction of the target. Could lead to the detection of a black hole at infinity. However, in terms of robotic arm positioning, LIDAR has a higher practical value. This paper will use the TFmini-Plus LiDAR system.

4) Binocular stereo imaging components

Binocular stereo vision is a method based on human vision. It uses binocular imaging technology to acquire 3D images of the target. The above data is then processed as a depth image to achieve 3D reconstruction. This method is not only intuitive, but also has high precision. The depth of field judgment. The same object is reconstructed in three dimensions and the images of two objects from different angles are obtained [7]. At the same time, the image will be deformed and deformed. Then the boundary frame deformation is converted into the target deformation by the deformation coefficient. The matrix of the camera's internal and external parameters is then converted into a forward image and its proportions are corrected. Finally, the effective image information is cut out. Binocular distance measurement has achieved good results, but the speed of the recognition of multiple objects will be limited. The OV9732's binocular camera can be used for positioning the robot.

C. System program design

The program of LPC2292 consists of main program, temperature acquisition subroutine, infrared signal acquisition subroutine, ultrasonic ranging subroutine, data processing and punching bag program, data communication program, EEPROM read and write program, external interrupt program and timer interrupt subroutine. The initialization of the system and the call of each subroutine are realized, and the test data is sent to DSP. The main programming process for the LCD 2292 is shown in Figure 5 (image cited in Symmetry 2022, 14(4), 777).

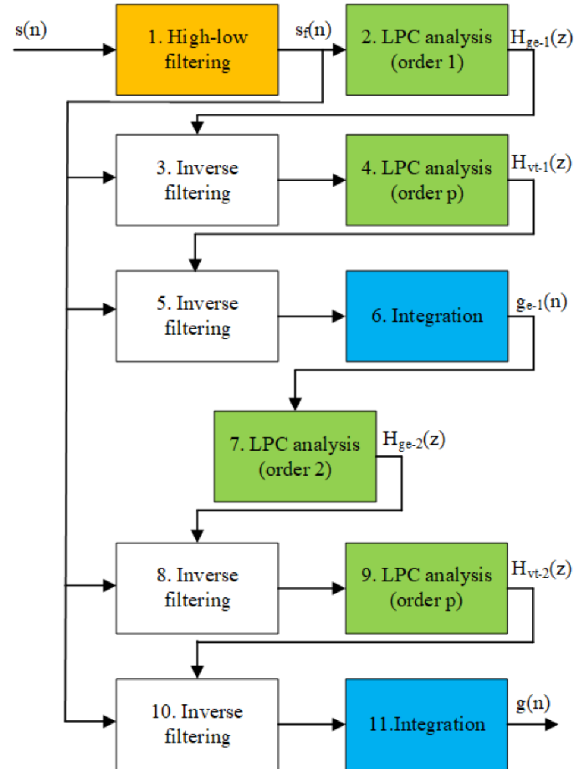


Fig. 5. Main program flow of LPC2292

III. SYSTEM ALGORITHM

In this paper, the measurement value $\{R_1, R_2, \dots, R_M\}$ of M sensor in M multi-sensor ranging system is regarded

as the identification frame ϕ , and the Mass function of the final synthesized evidence is regarded as the weight distribution function of A measurement value.

A. Accuracy judgment of measurement value

Suppose $|R_m - R_n| (m, n = 1, 2, \dots, M)$ is the distance between R_m and R_n , l_m represents the average distance between R_m and M measurements, and $\bar{l}$ represents the average distance between M measurements. If $l_m < \bar{l}$, then R_m belongs to the small deviation measurement set Λ_1 ; No side R_m belongs to the large deviation set Λ_2 .

B. Conversion of measured values to evidence

If $R_{m1} \subseteq \Lambda_1$, J of the measured values are located in $\Xi_{m1} = [R_{m1} - \bar{l}, R_{m1} + \bar{l}]$, then the distance between R_{m1} and the true value R is relatively close, and the Mass function of evidence e_{m1} obtained from R_{m1} is:

$$n_{m1}(X_j) = 1/J, \forall X_j \in \Xi_{m1} \quad (1)$$

That is, J measured values located in Ξ_{m1} of R_{m1} are regarded as focal elements, and all of them are assigned basic trust by $1/J$. The remaining $M - J$ measurements are assigned a basic trust of 0. If $R_{m2} \subseteq \Lambda_2, l_{\max} = \max\{R_m\} - \min\{R_m\}$, then R_{m2} is relatively far away from the true value R , and all M measured values are located in $\Xi_{m2} = [R_{m2} - l_{\max}, R_{m2} + l_{\max}]$, the Mass function of the evidence e_{m2} obtained from R_{m2} is:

$$n_{m2}(R_m) = 1/M, \forall R_m \subseteq \phi \quad (2)$$

That is, in evidence e_{m2} generated by R_{m2} , all measurements are assigned basic trust by $1/M$. The initial evidence $e_i (i = 1, 2, \dots, M)$ generated by the above M measurements may contain focal elements from both Λ_1 and Λ_2 . The deviation degree of M measured values is taken into account, and M initial evidence $e_i (m = 1, 2, \dots, M)$ is corrected respectively:

1) $\forall R_{j1}, R_{j2} \in \Lambda_1$, then the ratio of the basic trust distribution obtained by them is

$$\frac{n_i(R_{j1})}{n_i(R_{j2})} = \frac{l_{j2}}{l_{j1}} \quad (3)$$

2) $\forall R_j \in \Lambda_1$ and $\forall R_r \in \Lambda_2$, then the ratio of the basic trust distribution obtained by them is:

$$n_i(R_j)/n_i(R_r) = l_{\max}/l_j \quad (4)$$

A set of correction coefficients $\{\lambda_m\} (m = 1, 2, \dots, M)$ is obtained from equation (3-4). M pieces of evidence were weighted and normalized by using the correction coefficient, namely:

$$n'_i(R_m) = \lambda_m \cdot n_i(R_m) / \sum_{q=1}^M \lambda_q \cdot n_i(R_q) \quad (5)$$

C. Evidence Combination

M piece of evidence generated by equation (1-5) may have a high conflict, and the probability of supporting evidence conflict is allocated proportionally to each measurement value $R_m (m = 1, 2, \dots, M)$, and the combination formula is shown in equation (6).

$$n(R_m) = \prod_{i=1}^M n'_i(R_m) + \varepsilon \cdot \bar{n}_i(R_m) \quad (6)$$

Where ε is the conflict factor,

$$\varepsilon = 1 - \sum_{m=1}^M \prod_{i=1}^M n'_i(R_m) \quad (7)$$

$\bar{n}_i(R_m)$ is R_m average basic trust distribution across all evidence,

$$\bar{n}_i(R_m) = \frac{1}{M} \sum_{i=1}^M n_i(R_m) \quad (8)$$

The basic trust assignment $n(R_m)$ of R_m in synthetic evidence E is the weight obtained by R_m , then the fusion result is:

$$\hat{R} = \sum_{m=1}^M R_m \cdot n(R_m) \quad (9)$$

IV. EXPERIMENTAL RESULTS

The distance of the ultrasonic sensor is between 0.15 m and 10.7 m; The measurement accuracy is up to ± 3 decimeter. E18-B0 is a new infrared sensor, which is equipped with an induction switch that can adjust the sensing range according to the size of the power supply [8]. The maximum range is 5-60 cm. In this test, the maximum

detection range of the infrared detector is set within 50 cm. Set up a barrier 100 cm in front, and then move 10 cm at a time for one detection until it is 20 cm away from the barrier. Table I and II show the analysis results of the test results. R is the true value to be measured. $x_i(i=1,\cdots,6)$ indicates the measurement value measured by the sensor in area X_i . $n(R_i)(i=1,\cdots,6)$ represents the fusion weight

of the corresponding sensor. $\hat{R}$ represents the result of the fusion calculation of evidence theory. Δ represents $\hat{R}$ and R . $\bar{R}$ indicates the average value measured by each sensor. Λ represents the $\bar{R}$ and R . E is the relative error between Δ and Λ .

TABLE I. MEASUREMENT INFORMATION, FUSION WEIGHTS AND FUSION RESULTS OF SENSORS

R	R1	R2	R3	R4	R5	R6	n(R1)	n(R2)	n(R3)	n(R4)	n(R5)	n(R6)	R [^]
100	no	no	no	105.83	104.38	103.54	0	0	0	0.10	0.77	0.17	104.38
90	no	no	no	94.06	92.81	95.31	0	0	0	0.86	0.09	0.09	94.05
80	no	no	no	83.02	83.75	82.19	0	0	0	0.86	0.09	0.09	83.00
70	no	no	no	73.75	73.23	72.71	0	0	0	0.86	0.09	0.09	73.22
60	no	no	no	63.13	62.71	62.71	0	0	0	0.04	0.50	0.50	62.72
50	Yes	Yes	Yes	52.71	51.88	52.40	0.30	0.30	0.30	0.03	0.07	0.03	52.10
40	Yes	Yes	Yes	41.98	41.88	42.29	0	0	0	0.58	0.42	0.04	41.95
30	Yes	Yes	Yes	31.04	30.83	31.04	0	0	0	0.50	0.04	0.50	31.03
20	Yes	Yes	Yes	21.35	20.63	20.94	0	0	0	0.10	0.17	0.77	20.93

TABLE II. CALCULATION RESULTS AND RELATIVE ERRORS OF THE TWO METHODS

R	R [^]	$\bar{R}$	Δ	Λ	E
100	104.38	104.58	0.21%	0.42%	52.08%
90	94.05	94.06	0.33%	0.34%	3.13%
80	83.00	82.99	0.42%	0.44%	4.90%
70	73.22	73.23	0.44%	0.45%	2.40%
60	62.72	62.85	0.38%	0.58%	37.19%
50	52.10	52.33	0.03%	0.49%	97.50%
40	41.95	42.05	0.71%	0.96%	27.19%
30	31.03	30.97	0.72%	0.93%	23.44%
20	20.93	20.97	0.46%	0.70%	35.73%

It can be seen from the data in Table I that the measurement data of the ultrasonic sensor has high accuracy; As can be seen from the data in Table II, the measurement error after the fusion of evidence theory is reduced by 30.2% on average compared with that after the average method, which is obvious that the measured value after the fusion is closer to the true value.

V. CONCLUSION

This project intends to use multiple ranging sensors as the core and adopt the method of probability weight information fusion to realize the reliable ranging between multiple sensors. The scheme is suitable for positioning and avoiding of multi-scene and multi-function mobile robot. Improve its universality and robustness in unknown environments. The method of random analysis is used to integrate the valid data to make the final data more reliable. This reduces the impact of deviations caused by a single sensor. The research results of this project will lay the theoretical and technical foundation for the autonomous navigation and autonomous navigation of mobile robotic arms in the future.

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